

BA –UNIT 5

Introduction to Big Data, Master data management, Data mining on what kind of data, what kinds of patterns can be mines, which technologies are used, which kinds of applications are targeted, major issues in data mining.

Getting to know your data: Data Objects and Attribute Types, Basic Statistical descriptions of data, data visualization, Measuring data similarity and Dissimilarity.

What is big data analytics?

Big data analytics is the often complex process of examining big data to uncover information -- such as hidden patterns, correlations, market trends and customer preferences -- that can help organizations make informed business decisions. On a broad scale, data analytics technologies and techniques give organizations a way to analyze data sets and gather new information. Business intelligence (BI) queries answer basic questions about business operations and performance.

Big data analytics is a form of advanced analytics, which involve complex applications with elements such as predictive models, statistical algorithms and what-if analysis powered by analytics systems.

Big data analytics applications often include data from both internal systems and external sources, such as weather data or demographic data on consumers compiled by third-party information services providers. In addition, streaming analytics applications are becoming common in big data environments as users look to perform real-time analytics on data fed into Hadoop systems through stream processing engines, such as Spark, Flink and Storm.

Early big data systems were mostly deployed on premises, particularly in large organizations that collected, organized and analyzed massive amounts of data. But cloud platform vendors, such as Amazon Web Services (AWS), Google and Microsoft, have made it easier to set up and manage Hadoop clusters in the cloud. The same goes for Hadoop suppliers such as Cloudera, which supports the distribution of the big data framework on the AWS, Google and Microsoft Azure clouds. Users can now spin up clusters in the cloud, run them for as long as they need and then take them offline with usage based pricing that doesn't require ongoing software licenses.

Big data has become increasingly beneficial in supply chain analytics. Big supply chain analytics utilizes big data and quantitative methods to enhance decision-making processes across the supply chain. Specifically, big supply chain analytics expands data sets for increased analysis that goes beyond the traditional internal data found on enterprise resource planning (ERP) and supply chain management (SCM) systems. Also, big supply chain analytics implements highly effective statistical methods on new and existing data sources.

Why is big data analytics important?

Organizations can use big data analytics systems and software to make data-driven decisions that can improve business-related outcomes. The benefits may include more effective marketing, new revenue opportunities, customer personalization and improved operational efficiency. With an effective strategy, these benefits can provide competitive advantages over rivals.

How does big data analytics work?

Data analysts, data scientists, predictive modelers, statisticians and other analytics professionals collect, process, clean and analyze growing volumes of structured transaction data as well as other forms of data not used by conventional BI and analytics programs.

Here is an overview of the four steps of the big data analytics process:

Data professionals **collect** data from a variety of different sources. Often, it is a mix of semistructured and unstructured data. While each organization will use different data streams, some common sources include:

- internet clickstream data;
- web server logs;
- cloud applications;
- mobile applications;
- social media content;
- text from customer emails and survey responses;
- mobile phone records; and
- machine data captured by sensors connected to the internet of things (IoT).

Data is prepared and **processed**. After data is collected and stored in a data warehouse or data lake, data professionals must organize, configure and partition the data properly for analytical queries. Thorough data preparation and processing makes for higher performance from analytical queries. Data is **cleansed** to improve its quality. Data professionals scrub the data using scripting tools or data quality software. They look for any errors or inconsistencies, such as duplications or formatting mistakes, and organize and tidy up the data.

The collected, processed and cleaned data is **analyzed** with analytics software. This includes tools for:

- data mining, which sifts through data sets in search of patterns and relationships
- predictive analytics, which builds models to forecast customer behavior and other future actions, scenarios and trends
- machine learning, which taps various algorithms to analyze large data sets
- deep learning, which is a more advanced offshoot of machine learning
- text mining and statistical analysis software
- artificial intelligence (AI)
- mainstream business intelligence software
- data visualization tools

Key big data analytics technologies and tools

Many different types of tools and technologies are used to support big data analytics processes. Common technologies and tools used to enable big data analytics processes include: **Hadoop**, which is an open source framework for storing and processing big data sets. Hadoop can handle large amounts of structured and unstructured data.

Predictive analytics hardware and software, which process large amounts of complex data, and use machine learning and statistical algorithms to make predictions about future event outcomes. Organizations use predictive analytics tools for fraud detection, marketing, risk assessment and operations.

Stream analytics tools, which are used to filter, aggregate and analyze big data that may be stored in many different formats or platforms.

Distributed storage data, which is replicated, generally on a non-relational database. This can be as a measure against independent node failures, lost or corrupted big data, or to provide low-latency access.

NoSQL databases, which are non-relational data management systems that are useful when working with large sets of distributed data. They do not require a fixed schema, which makes them ideal for raw and unstructured data.

A data lake is a large storage repository that holds native-format raw data until it is needed. Data lakes use a flat architecture.

A data warehouse, which is a repository that stores large amounts of data collected by different sources. Data warehouses typically store data using predefined schemas.

Knowledge discovery/big data mining tools, which enable businesses to mine large amounts of structured and unstructured big data.

In-memory data fabric, which distributes large amounts of data across system memory resources. This helps provide low latency for data access and processing.

Data virtualization, which enables data access without technical restrictions.

Data integration software, which enables big data to be streamlined across different platforms, including Apache, Hadoop, MongoDB and Amazon EMR.

Data quality software, which cleanses and enriches large data sets.

Data preprocessing software, which prepares data for further analysis. Data is formatted and unstructured data is cleansed.

Spark, which is an open source cluster computing framework used for batch and stream data processing.

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Big data analytics uses and examples

Here are some examples of how big data analytics can be used to help organizations: **Customer acquisition and retention.** Consumer data can help the marketing efforts of companies, which can act on trends to increase customer satisfaction. For example, personalization engines for Amazon, Netflix and Spotify can provide improved customer experiences and create customer loyalty. **Targeted ads.** Personalization data from sources such as past purchases, interaction patterns and product page viewing histories can help generate compelling targeted ad campaigns for users on the individual level and on a larger scale.

Product development. Big data analytics can provide insights to inform about product viability, development decisions, progress measurement and steer improvements in the direction of what fits a business' customers.

Price optimization. Retailers may opt for pricing models that use and model data from a variety of data sources to maximize revenues.

Supply chain and channel analytics. Predictive analytical models can help with preemptive replenishment, B2B supplier networks, inventory management, route optimizations and the notification of potential delays to deliveries.

Risk management. Big data analytics can identify new risks from data patterns for effective risk management strategies.

Improved decision-making. Insights business users extract from relevant data can help organizations make quicker and better decisions.

Big data analytics benefits

The benefits of using big data analytics include:

Quickly analyzing large amounts of data from different sources, in many different formats and types.

Rapidly making better-informed decisions for effective strategizing, which can benefit and improve the supply chain, operations and other areas of strategic decision-making.

Cost savings, which can result from new business process efficiencies and optimizations. A better understanding of customer needs, behavior and sentiment, which can lead to better marketing insights, as well as provide information for product development.

Improved, better informed risk management strategies that draw from large sample sizes of data.

Big data analytics challenges

Despite the wide-reaching benefits that come with using big data analytics, its use also comes with challenges:

Accessibility of data. With larger amounts of data, storage and processing become more complicated. Big data should be stored and maintained properly to ensure it can be used by less experienced data scientists and analysts.

Data quality maintenance. With high volumes of data coming in from a variety of sources and in different formats, data quality management for big data requires significant time, effort and resources to properly maintain it.

Data security. The complexity of big data systems presents unique security challenges. Properly addressing security concerns within such a complicated big data ecosystem can be a complex undertaking.

Choosing the right tools. Selecting from the vast array of big data analytics tools and platforms available on the market can be confusing, so organizations must know how to pick the best tool that aligns with users' needs and infrastructure.

With a potential lack of internal analytics skills and the high cost of hiring experienced data scientists and engineers, some organizations are finding it hard to fill the gaps.

History and growth of big data analytics

The term big data was first used to refer to increasing data volumes in the mid-1990s. In 2001, Doug Laney, then an analyst at consultancy Meta Group Inc., expanded the definition of big data. This expansion described the increasing:

Volume of data being stored and used by organizations;

Variety of data being generated by organizations; and

Velocity, or speed, in which that data was being created and updated.

Those three factors became known as the 3Vs of big data. Gartner popularized this concept after acquiring Meta Group and hiring Laney in 2005.

Another significant development in the history of big data was the launch of the Hadoop distributed processing framework. Hadoop was launched as an Apache open source project in 2006. This planted the seeds for a clustered platform built on top of commodity hardware and that could run big data applications. The Hadoop framework of software tools is widely used for managing big data.

By 2011, big data analytics began to take a firm hold in organizations and the public eye, along with Hadoop and various related big data technologies.

Initially, as the Hadoop ecosystem took shape and started to mature, big data applications were primarily used by large internet and e-commerce companies such as Yahoo, Google and Facebook, as well as analytics and marketing services providers.

More recently, a broader variety of users have embraced big data analytics as a key technology driving digital transformation. Users include retailers, financial services firms, insurers, healthcare organizations, manufacturers, energy companies and other enterprises.

This was last updated in December 2021



What are the 5 V's of Big Data?

Big data is a collection of data from many different sources and is often describe by five characteristics: **volume, value, variety, velocity, and veracity.**

Volume: The size and amounts of big data that companies manage and analyze. **Value:** the most important “V” from the perspective of the business, the value of big data usually comes from insight discovery and pattern recognition that lead to more effective operations, stronger customer relationships and other clear and quantifiable business benefits

Variety: The diversity and range of different data types, including unstructured data, semi-structured data and raw data

Unstructured data types

 Text files and documents	 Server, website and application logs	 Sensor data	 Images
 Video files	 Audio files	 Emails	 Social media data

Velocity: The speed at which companies receive, store and manage data – e.g., the specific number of social media posts or search queries received within a day, hour or other unit of time **Veracity:** The “truth” or accuracy of data and information assets, which often determines executive level confidence

The 5 V's of big data (velocity, volume, value, variety and veracity) are the five main and innate characteristics of big data. Knowing the 5 V's allows data scientists to derive more value from their data while also allowing the scientists' organization to become more customer-centric. In the early part of this century, big data was only talked about in terms of the three V's -- volume, velocity and variety. Over time, two more V's (value and veracity) have been added to help data scientists be more effective in articulating and communicating the important characteristics of big data. The number five mirrors the five basic questions every news article should answer. However, it is not specifically required that organizations follow data guideline over the other.

The history of big data

Although the concept of big data itself is relatively new, the origins of large data sets go back to the 1960s and '70s when the world of data was just getting started with the first data centers and the development of the relational database.

Around 2005, people began to realize just how much data users generated through Facebook, YouTube, and other online services. Hadoop (an open-source framework created specifically to store and analyze big data sets) was developed that same year. NoSQL also began to gain popularity during this time. The development of open-source frameworks, such as Hadoop (and more recently, Spark) was essential for the growth of big data because they make big data easier to work with and cheaper to store. In the years since then, the volume of big data has skyrocketed. Users are still generating huge amounts of data—but it's not just humans who are doing it.

With the advent of the Internet of Things (IoT), more objects and devices are connected to the internet, gathering data on customer usage patterns and product performance. The emergence of machine learning has produced still more data.

While big data has come far, its usefulness is only just beginning. Cloud computing has expanded big data possibilities even further. The cloud offers truly elastic scalability, where developers can simply spin

up ad hoc clusters to test a subset of data. And graph databases are becoming increasingly important as well, with their ability to display massive amounts of data in a way that makes analytics fast and comprehensive.

Big data benefits:

Big data makes it possible for you to gain more complete answers because you have more information. More complete answers mean more confidence in the data—which means a completely different approach to tackling problems.

Big data use cases

Big data can help you address a range of business activities, from customer experience to analytics. Here are just a few.

Product development	Companies like Netflix and Procter & Gamble use big data to anticipate customer demand. They build predictive models for new products and services by classifying key attributes of past and current products or services and modeling the relationship between those attributes and the commercial success of the offerings. In addition, P&G uses data and analytics from focus groups, social media, test markets, and early store rollouts to plan, produce, and launch new products.
Predictive maintenance	Factors that can predict mechanical failures may be deeply buried in structured data, such as the year, make, and model of equipment, as well as in unstructured data that covers

	millions of log entries, sensor data, error messages, and engine temperature. By analyzing these indications of potential issues before the problems happen, organizations can deploy maintenance more cost effectively and maximize parts and equipment uptime.
Customer experience	The race for customers is on. A clearer view of customer experience is more possible now than ever before. Big data enables you to gather data from social media, web visits, call logs, and other sources to improve the interaction experience and maximize the value delivered. Start delivering personalized offers, reduce customer churn, and handle issues proactively.
Fraud and compliance	When it comes to security, it's not just a few rogue hackers—you're up against entire expert teams. Security landscapes and compliance requirements are constantly evolving. Big data helps you identify patterns in data that indicate fraud and aggregate large volumes of information to make regulatory reporting much faster.
Machine learning	Machine learning is a hot topic right now. And data—specifically big data—is one of the reasons why. We are now able to teach machines instead of program them. The availability of big data to train machine learning models

	makes that possible.
Operational efficiency	Operational efficiency may not always make the news, but it's an area in which big data is having the most impact. With big data, you can analyze and assess production, customer feedback and returns, and other factors to reduce outages and anticipate future demands. Big data can also be used to improve decision-making in line with current market demand.
Drive innovation	Big data can help you innovate by studying interdependencies among humans, institutions, entities, and process and then determining new ways to use those insights. Use data insights to improve decisions about financial and planning considerations. Examine trends and what customers want to deliver new products and services. Implement dynamic pricing. There are endless possibilities.

Big data best practices

To help you on your big data journey, we've put together some key best practices for you to keep in mind. Here are our guidelines for building a successful big data foundation.

Align big data with specific business goals

More extensive data sets enable you to make new discoveries. To that end, it is important to base new investments in skills, organization, or infrastructure with a strong business-driven context to guarantee ongoing project investments and funding. To determine if you are on the right track, ask how big data supports and enables your top business and IT priorities. Examples include understanding how to filter web logs to

Top payoff is aligning unstructured with structured data

Ease skills shortage with standards and governance

Optimize knowledge transfer with a center of excellence

Plan your discovery lab for performance understand ecommerce behavior, deriving sentiment from social media and customer support interactions, and understanding statistical correlation methods and their relevance for customer, product, manufacturing, and engineering data.

One of the biggest obstacles to benefiting from your

investment in big data is a skills shortage. You can mitigate this risk by ensuring that big data technologies, considerations, and decisions are added to your IT governance program. Standardizing your approach will allow you to manage costs and leverage resources. Organizations implementing big data solutions and strategies should assess their skill requirements early and often and should proactively identify any potential skill gaps. These can be addressed by training/cross-training existing resources, hiring new resources, and leveraging consulting firms. Use a center of excellence approach to share knowledge, control oversight, and manage project communications. Whether big data is a new or expanding investment, the soft and hard costs can be shared across the enterprise. Leveraging this approach can help increase big data capabilities and overall information architecture maturity in a more structured and systematic way. It is certainly valuable to analyze big data on its own. But you can bring even greater business insights by connecting and integrating low density big data with the structured data you are already using today. Whether you are capturing customer, product,

equipment, or environmental big data, the goal is to add more relevant data points to your core master and analytical summaries, leading to better conclusions. For example, there is a difference in distinguishing all customer sentiment from that of only your best customers. Which is why many see big data as an integral extension of their existing business intelligence capabilities, data warehousing platform, and information architecture.

Keep in mind that the big data analytical processes and models can be both human- and machine-based. Big data analytical capabilities include statistics, spatial analysis, semantics, interactive discovery, and visualization. Using analytical models, you can correlate different types and sources of data to make associations and meaningful discoveries.

Discovering meaning in your data is not always straightforward. Sometimes we don't even know what we're looking for. That's expected.

Management and IT needs to support this "lack of direction" or "lack of clear requirement."

At the same time, it's important for analysts and data scientists to work closely with the business to understand key business knowledge gaps and requirements. To accommodate the

interactive exploration of data and the experimentation of statistical algorithms, you need high-performance work areas.

Be sure that sandbox environments have the support they need—and are properly governed.

Align with the cloud operating model Big data processes and users require access to a broad array of resources for both iterative experimentation and running production jobs. A big data solution includes all data realms including transactions, master data, reference data, and summarized data. Analytical sandboxes should be created on demand. Resource management is critical to ensure control of the entire data flow including pre- and post-processing, integration, in-database summarization, and analytical modeling. A well-planned private and public cloud provisioning and security strategy plays an integral role in supporting these changing requirements.

Following are some of the Big Data examples

The **New York Stock Exchange** is an example of Big Data that generates about *one terabyte* of new trade data per day.



Social Media

The statistic shows that *500+terabytes* of new data get ingested into the databases of social media site **Facebook**, every day. This data is mainly generated in terms of photo and video uploads, message exchanges, putting comments etc.



A single **Jet engine** can generate *10+terabytes* of data in *30 minutes* of flight time. With many thousand flights per day, generation of data reaches up to many *Petabytes*.



Types Of Big Data

Following are the types of Big Data:

Structured

Any data that can be stored, accessed and processed in the form of fixed format is termed as a 'structured' data. Over the period of time, talent in computer science has achieved greater success in developing techniques for working with such kind of data (where the format is well known in advance) and also deriving value out of it. However, nowadays, we are foreseeing issues when a size of such data grows to a huge extent, typical sizes are being in the rage of multiple zettabytes.

Do you know? 1021 bytes equal to 1 zettabyte or one billion terabytes forms a zettabyte. Looking at these figures one can easily understand why the name Big Data is given and imagine the challenges involved in its storage and processing.

Do you know? Data stored in a relational database management system is one example of a ‘structured’ data.

Examples Of Structured Data

An ‘Employee’ table in a database is an example of Structured Data

EmpId	EmpName	Gender	Department	Salary_In_Lacs
-------	---------	--------	------------	----------------

2365 Rajesh Kulkarni Male Finance 650000

3398	Pratibha Joshi	Female	Admin	650000
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7465 Shushil Roy Male Admin 500000

7500	Shubhojit Das	Male	Finance	500000
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7699 Priya Sane Female Finance 550000

Unstructured

Any data with unknown form or the structure is classified as unstructured data. In addition to the size being huge, un-structured data poses multiple challenges in terms of its processing for deriving value out of it. A typical example of unstructured data is a heterogeneous data source containing a combination of simple text files, images, videos etc. Now day organizations have wealth of data available with them but unfortunately, they don’t know how to derive value out of it since this data is in its raw form or unstructured format.

Examples Of Un-structured Data

The output returned by ‘Google Search’

The screenshot displays a Google search interface with the query 'hadoop big data'. The search results include several links to websites like IBM, Wandisco, and Simplilearn, which offer Hadoop training and enterprise solutions. On the right side, there is a 'Shop for hadoop big data on Google' section, which is a sponsored advertisement. This section lists various books related to Hadoop and Big Data, such as 'Big Data Analytics', 'Hadoop Beginner's Guide', and 'Hadoop The Definitive Guide', along with their prices and where they can be purchased (Amazon.in or Flipkart).

Example Of Un-structured Data

Semi-structured

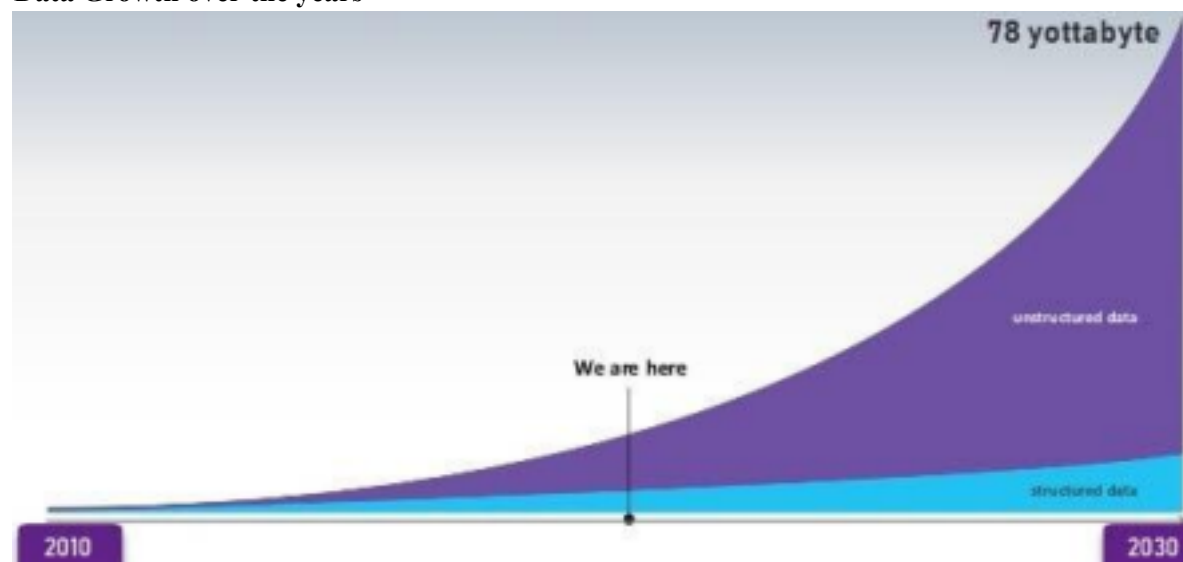
Semi-structured data can contain both the forms of data. We can see semi-structured data as a structured in form but it is actually not defined with e.g. a table definition in relational DBMS. Example of semi-structured data is a data represented in an XML file.

Examples Of Semi-structured Data

Personal data stored in an XML file-

```
<rec><name>Prashant Rao</name><sex>Male</sex><age>35</age></rec>
<rec><name>Seema R.</name><sex>Female</sex><age>41</age></rec>
<rec><name>Satish Mane</name><sex>Male</sex><age>29</age></rec>
<rec><name>Subrato Roy</name><sex>Male</sex><age>26</age></rec>
<rec><name>Jeremiah J.</name><sex>Male</sex><age>35</age></rec>
```

Data Growth over the years



Master Data Management (MDM) & Big Data |

In the internet's infancy, before search engines and browsers, you had to know exactly where to look to find the information you wanted.

With the emergence of search engines, the entire wealth of information became available at our fingertips just by inserting a keyword. Google's most significant achievement involved making sense of big data served up from over a billion websites globally according to user preferences, location and history. Google simply and elegantly integrated the world wide web to deliver information with speed and relevance for the user.

Today, big data does not only exist on the internet but within single companies and organizations who need to capitalize on that data. Searchability no longer suffices; businesses need advanced data analytics that can both accurately describe and predict behavior and movements. That's the power of analytics – to discover new correlations of different data sources and blend data so that meaningful patterns of information emerge.



Master Data Management (MDM) as the catalyst

Master data management (MDM) is a technology-enabled discipline in which business and IT work together to ensure the uniformity, accuracy, stewardship, semantic consistency and accountability of the enterprise's official shared master data assets. Master data is the consistent and uniform set of identifiers and extended attributes that describes the core entities of the enterprise including customers, prospects, citizens, suppliers, sites, hierarchies and chart of accounts.

In the business industry, master data management is a cutting-edge method used in defining and managing the critical data of an organization to provide and integrate data to a single point of reference. This is accomplished by identifying the different types of data which is then followed by analysis and accurate management thereof, and finally, storage and distribution throughout an organization to ensure a common understanding.

Master data management is the natural application hub for bringing together this diverse array of data in a way that provides structure and data confidence.

The reason: master data represents your most essential business assets, including products, customers, locations, suppliers and vendors, digital and physical assets, employees and more. Each of these data domains and their interrelationships can have hundreds of attributes and billions of records that need to be clean and consolidated. High-quality master data is the foundation to this – as well as any digital transformation initiative.

For many retailers, manufacturing companies and financial institutions, a master data management solution is a critical data management tool because it streamlines master data for running operations and provides reliable data quality for analytics across both structured and unstructured data sources.

Having a powerful analytics engine embedded in your master data management solution enables you to derive new kinds of insights from these sources when correlating them with your master data. The value of analytics comes from being accessible and reliable.

Data analytics, when embedded into the master data management solution, further fuels the multi domain efforts. With embedded data analytics you can blend other types of data into the master data view and discover new connections in the landscape of big data.

Data analytics shows its full potential when it is not limited to a certain type of data but instead integrates

data from a multitude of sources: IoT, transactions, social media, production workflows to name a few. Each of these sources creates data that flows into various data repositories and data lakes where you run the risk of having valuable data lying idly around because it is not easily accessible to your organization.

Benefits of master data management

A master data management solution facilitates developing a single source of truth for any desired data domain. The single source of truth, often labelled as the “golden record,” lays the foundation for:

Streamlining onboarding processes through data syndication

Creating omnichannel and customer-centric experiences

Enhancing collaboration through enterprise solutions

Agile business management capable of responding quickly to market trends

Master data management helps to break down and connect data silos and prepares a true enterprise solution with global potential by enabling a rich ecosystem of interconnected data and processes.

Today, enterprises are becoming more customer-centric and are trying to know more and more about their customer preferences by collecting all kinds of data from available sources. These data, essentially termed as ‘big data’, typically encompasses large volumes of texts and other forms of unstructured behavioral data from a variety of sources. Master data management (MDM) primarily revolves around the creation of a trusted source of highly structured data throughout an enterprise. Data management analysts see a better future for enterprises through better insight into their customers, when they use MDM and big data in tandem. Let us see what is driving this.

Big data tools enable the analysis of massive volumes of data from various sources, however, the real challenge is:

- Being really able to relate this data with customers
- Actually knowing who the customers are
- Knowing your best customers
- Knowing how your trusted customers are reacting to your product or service

While big data is certainly an enabler in the direction of customer centricity, it alone does not serve the purpose. What is required is a combination of a strong MDM along with big data to outpace challenges and realize this objective.

Let us see how enterprises can leverage this combination of MDM and big data to build customer centric strategies:

To address the challenges posed from using just big data, there comes the need for a strong enterprise level MDM initiative that has a bigger role to play in aggregating useful information from pools of data and then matching it against the master data spread across an enterprise’s transactional systems. Large volume of processed data start making more sense only when it is tagged and related to the master data entities of the organization.

Enterprises are starting to see the benefits big data and MDM together could bring. According to a survey result by The Information Difference Ltd. an MDM consulting and Research Company, 67% of survey respondents saw MDM driving big data, rather than the other way around, with just 17% seeing big data producing new master data, including the ability to use master data to automatically detect customer names in sets of big data. The most popular choice was for existing MDM data to help drive big data searches.

Essentially, the foundation needs to be laid by master data across a variety of sources that big data brings along. Enterprises will need to move from a single customer record to a single customer profile which will provide an impetus to their business goals. While analyzing big data to unearth findings and insights is important, doing that on clean master data across locations, products, customers, cost centers, plants and other important master data domains will provide the right direction.

So for those who want to invest in big data, pause a while and think again. Are you ready for relating data with master data entity in your enterprise? Will a big data strategy alone help you to understand customers better or would it be a perfect combination of MDM and big data?

What Is Data Mining: Definition, Benefits



We are living in an information-rich, data-driven world. While it's comforting to know there's a plethora of readily available knowledge, the sheer volume creates challenges. The more information available, the longer it can find the useful insights you need.

That's why today we're discussing data mining. We'll be exploring all aspects of data mining, including what it means, its stages, data mining techniques, the benefits it offers, data mining tools, and more.

What is Data Mining?

Typically, when someone talks about "mining," it involves people wearing helmets with lamps attached to them, digging underground for natural resources. And while it could be funny picturing guys in tunnels mining for batches of zeroes and ones, that doesn't exactly answer "what is data mining." Data mining is the process of analyzing enormous amounts of information and datasets, extracting (or "mining") useful intelligence to help organizations solve problems, predict trends, mitigate risks, and find new opportunities. Data mining is like actual mining because, in both cases, the miners are sifting through mountains of material to find valuable resources and elements.

Data mining also includes establishing relationships and finding patterns, anomalies, and correlations to tackle issues, creating actionable information in the process. Data mining is a wide-ranging and varied process that includes many different components, some of which are even confused for data mining itself. For instance, statistics is a portion of the overall data mining process. Additionally, both data mining and machine learning fall under the general heading of data science, and though they have some similarities, each process works with data in a different way. If you want to know more about their relationship, read up on data mining vs. machine learning.

Data mining is sometimes called Knowledge Discovery in Data, or KDD.

Data Mining History

For millennia, people have excavated places to find hidden mysteries. "Knowledge discovery in databases" refers to the act of sifting through data to uncover hidden relationships and forecast future trends. In the 1990s, the phrase "data mining" was invented. Data mining emerged from the convergence of three scientific disciplines: artificial intelligence, machine learning, and statistics.

Artificial intelligence is the human-like intelligence demonstrated by software and machines, machine learning is the term used to describe algorithms that can learn from data to create predictions, and statistics is the numerical study of data correlations.

Data mining takes advantage of big data's infinite possibilities and inexpensive processing power. Processing power and speed have grown significantly in the recent decade, allowing the globe to undertake rapid, easy, and automated data analysis.

Data Mining Steps

When asking "what is data mining," let's break it down into the steps data scientists and analysts take

when tackling a data mining project.

1. Understand Business

What is the company's current situation, the project's objectives, and what defines success?

2. Understand the Data

Figure out what kind of data is needed to solve the issue, and then collect it from the proper sources.

3. Prepare the Data

Resolve data quality problems like duplicate, missing, or corrupted data, then prepare the data in a format suitable to resolve the business problem.

4. Model the Data

Employ algorithms to ascertain data patterns. Data scientists create, test, and evaluate the model. 5.

Evaluate the Data

Decide whether and how effective the results delivered by a particular model will help meet the business goal or remedy the problem. Sometimes there's an iterative phase for finding the best algorithm, especially if the data scientists don't get it quite right the first time. There may be some data mining algorithms shopping around.

6. Deploy the Solution

Give the results of the project to the people in charge of making decisions.

Examples of Data Mining

The following are a few real-world examples of data:

Shopping Market Analysis

In the shopping market, there is a big quantity of data, and the user must manage enormous amounts of data using various patterns. To do the study, market basket analysis is a modeling approach. Market basket analysis is basically a modeling approach that is based on the notion that if you purchase one set of products, you're more likely to purchase another set of items. This strategy may help a retailer understand a buyer's purchasing habits. Using differential analysis, data from different businesses and consumers from different demographic groups may be compared. **Weather Forecasting Analysis**

For prediction, weather forecasting systems rely on massive amounts of historical data. Because massive amounts of data are being processed, the appropriate data mining approach must be used. **Stock Market Analysis**

In the stock market, there is a massive amount of data to be analyzed. As a result, data mining techniques are utilized to model such data in order to do the analysis.

Intrusion Detection

Well, data mining can assist to enhance intrusion detection by focusing on anomaly detection. It assists an analyst in distinguishing between unusual network activity and normal network activity. **Fraud Detection**

Traditional techniques of fraud detection are time-consuming and difficult due to the amount of data.

Data mining aids in the discovery of relevant patterns and the transformation of data into information.

Surveillance

Well, video surveillance is utilized practically everywhere in everyday life for security perception. Because we must deal with a huge volume of acquired data, data mining is employed in video surveillance.

Financial Banking

With each new transaction in computerized banking, a massive amount of data is expected to be created. By identifying patterns, causalities, and correlations in corporate data, data mining may help solve business challenges in banking and finance.

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What Are the Benefits of Data Mining?

Since we live and work in a data-centric world, it's essential to get as many advantages as possible. Data mining provides us with the means of resolving problems and issues in this challenging information age. Data mining benefits include:

- It helps companies gather reliable information
- It's an efficient, cost-effective solution compared to other data applications
- It helps businesses make profitable production and operational adjustments
- Data mining uses both new and legacy systems
- It helps businesses make informed decisions
- It helps detect credit risks and fraud
- It helps data scientists easily analyze enormous amounts of data quickly

Data scientists can use the information to detect fraud, build risk models, and improve product safety. It helps data scientists quickly initiate automated predictions of behaviors and trends and discover hidden patterns.

Challenges of Implementation in Data Mining

Because data handling technology is always improving, leaders confront additional obstacles in addition to scalability and automation, as mentioned below:

Distributed Data

Real-world data saved on several platforms, such as databases, individual systems, or the Internet, cannot be transferred to a centralized repository. Regional offices may have their own servers to store data, but storing data from all offices centrally will be impossible. As a result, tools and algorithms for mining dispersed data must be created for data mining.

Complex Data

It takes a long time and money to process big amounts of complicated data. Data in the real world is structured, unstructured, semi-structured, and heterogeneous forms, including multimedia such as photos, music, video, natural language text, time series, natural, and so on, making it challenging to extract essential information from many sources in LAN and WAN.

Domain Knowledge

It is simpler to dig some information with domain expertise, without which collecting useful information from data might be tough.

Data Visualization

The first interaction that presents the result correctly to the client is data visualization. The information is conveyed with unique relevance based on its intended use. However, it is difficult to accurately address the information to the end-user. To make the information relevant, effective output information, input data, and complicated data perception methods must be used.

Incomplete Data

Large data amounts might be imprecise or unreliable owing to measurement equipment problems. Customers that refuse to disclose their personal information may result in incomplete data, which may be updated owing to system failures, resulting in noisy data, making the data mining procedure difficult.

Security and Privacy

Decision-making techniques necessitate security through data exchange for people, organizations, and the government. Private and sensitive information about individuals is gathered for customer profiles in order to better understand user activity trends. Illegal access and the confidentiality of the information are significant issues here.

Higher Costs

The expenses linked with purchasing and maintaining strong servers, software, and hardware for handling massive amounts of data might be too expensive.

Performance Issues

The performance of a data mining system is determined by the methods and techniques utilized, which might have an impact on data mining performance. Large database volumes, data flow, and data mining challenges can all contribute to the development of parallel and distributed data mining methods.

Data Mining Prerequisites

Data mining necessitates an understanding of arithmetic and statistics, programming, business principles, and communication. To begin studying data analysis, you must have knowledge in the following areas:

- Linear Algebra
- Artificial Intelligence
- Machine Learning
- Statistical Analysis
- Data Structures and Algorithms
- Data Retrieval and Database
- Problem-solving Ability

The Future of Data Mining

The future of data mining is bright, as data volumes continue to grow. Mining techniques have changed as a result of technological advancements, as have systems that extract useful information from data. Previously, only companies such as NASA could utilize their supercomputers to examine data since the expense of storing and calculating data was prohibitively expensive.

Companies are now experimenting with machine learning, artificial intelligence, and deep learning on cloud-based data lakes.

The Internet of Things and wearable technologies have transformed people and gadgets into data generating machines capable of producing infinite knowledge about individuals and organizations. This is how businesses can gather, store, and analyze massive amounts of data.

Cloud-based analytics solutions will make it easier and more cost-effective for businesses to access huge amounts of data and processing power. Cloud computing enables businesses to swiftly receive and act on data from sales, marketing, Internet, manufacturing, and inventory systems, among other sources, in order to enhance their bottom line.

What Kinds of Data Mining Tools Are Out There?

As engineers are fond of saying, “Use the right tool for the right job.” Here is a selection of tools and techniques that provide data analysts with diverse data mining functionalities.

Artificial Intelligence

AI systems perform analytical functions that mimic human intelligence, such as learning, planning, problem-solving, and reasoning.

Association Rule Learning

This toolset, also called market basket analysis, searches for relationships among dataset variables. For example, association rule learning can determine which products are frequently purchased together (e.g., a smartphone and a protective case).

Clustering

This process partitions datasets into a set of meaningful sub-classes, known as clusters. The process helps users understand the natural structure or grouping within the data.

Classification

This technique assigns particular items in a dataset to different target categories or classes. The goal is to develop accurate predictions within the target class for each case in the data. **Data Analytics**

The data analytics process enables professionals to evaluate digital information and turn it into useful business intelligence.

Data Cleansing and Preparation

This technique transforms the data into a form optimal for further analysis and processing. Preparation includes activities such as identifying and removing errors and missing or duplicate data. **Data**

Warehousing

Data warehousing consists of an extensive collection of business data that businesses use to help them make decisions. Warehousing is a fundamental and necessary component of most large-scale data mining efforts.

Machine Learning

Related to the AI technique mentioned earlier, machine learning is a computer programming technique that employs statistical probabilities to provide computers with the ability to learn without human intervention or being manually programmed.

Regression

The regression technique predicts a range of numeric values in categories such as sales, stock prices, or even temperature. The ranges are based on the information found in a particular data set. Two specific tools need mentioning.

R. This language is an open-source tool used for graphics and statistical computing. It provides analysts with a wide selection of statistical tests, classification and graphical techniques, and time-series analysis.

Oracle Data Mining (ODM). This tool is a module of the Oracle Advanced Analytics Database. It helps data analysts make predictions and generate detailed insights. Analysts use ODM to predict customer behavior, develop customer profiles, and identify cross-selling opportunities.

In our learning about what is data mining, let us now look into the applications.

Data Mining Applications

Data mining is a useful and versatile tool for today's competitive businesses. Here are some data mining examples, showing a broad range of applications.

Banks

Data mining helps banks work with credit ratings and anti-fraud systems, analyzing customer financial data, purchasing transactions, and card transactions. Data mining also helps banks better understand their customers' online habits and preferences, which helps when designing a new marketing campaign.

Healthcare

Data mining helps doctors create more accurate diagnoses by bringing together every patient's medical history, physical examination results, medications, and treatment patterns. Mining also helps fight fraud and waste and bring about a more cost-effective health resource management strategy.

Marketing

If there was ever an application that benefitted from data mining, it's marketing! After all, marketing's heart and soul is all about targeting customers effectively for maximum results. Of course, the best way to target your audience is to know as much about them as possible. Data mining helps bring together data on age, gender, tastes, income level, location, and spending habits to create more effective personalized loyalty campaigns. Data marketing can even predict which customers will more likely unsubscribe to a mailing list or other related service. Armed with that information, companies can take steps to retain those customers before they get the chance to leave!

Retail

The world of retail and marketing go hand-in-hand, but the former still warrants its separate listing. Retail stores and supermarkets can use purchasing patterns to narrow down product associations and determine which items should be stocked in the store and where they should go. Data mining also pinpoints which campaigns get the most response.

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What Kinds of Patterns Can Data Mining Discover?

Using the most relevant data (which may come from organizational databases or may be obtained from outside sources), data mining builds models to identify patterns among the attributes (i.e., variables or characteristics) that exist in a data set. Models are usually the mathematical representations (simple linear correlations and/or complex highly nonlinear relationships) that identify the relationships among the attributes of the objects (e.g., customers) described in the data set. Some of these patterns are explanatory (explaining the interrelationships and affinities among the attributes), whereas others are predictive (projecting future values of certain attributes). In general, data mining seeks to identify three major types of patterns:

Associations find commonly co-occurring groupings of things, such as "beers and diapers" or "bread and butter" commonly purchased and observed together in a shopping cart (i.e., market-basket analysis).

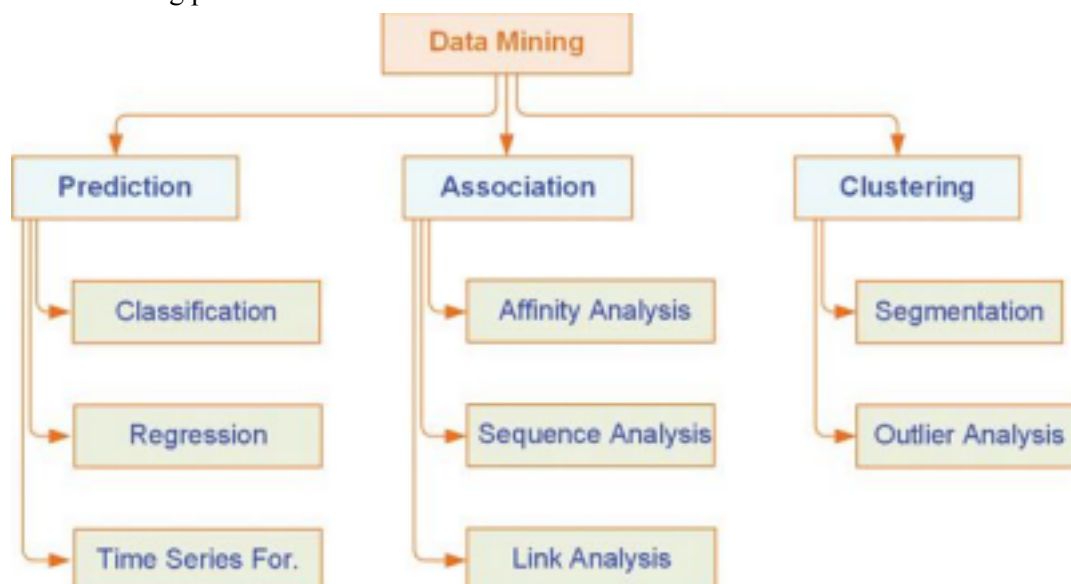
Another type of association pattern captures the sequences of things. These sequential relationships can discover time-ordered events, such as predicting that an existing banking customer who already has a checking account will open a savings account followed by an investment account within a year.

Predictions tell the nature of future occurrences of certain events based on what has happened in the past, such as predicting the winner of the Super Bowl or forecasting the absolute temperature on a particular

day.

Clusters identify natural groupings of things based on their known characteristics, such as assigning customers in different segments based on their demographics and past purchase behaviors. These types of patterns have been manually extracted from data by humans for centuries, but the increasing volume of data in modern times has created a need for more automatic approaches. As data sets have grown in size and complexity, direct manual data analysis has increasingly been augmented with indirect, automatic data processing tools that use sophisticated methodologies, methods, and algorithms. The manifestation of such evolution of automated and semi-automated means of processing large data sets is now commonly referred to as data mining.

As mentioned earlier, generally speaking, data mining tasks and patterns can be classified into three main categories: **prediction**, **association**, and **clustering**. Based on the way in which the patterns are extracted from the historical data, the learning algorithms of data mining methods can be classified as either supervised or unsupervised. With supervised learning algorithms, the training data includes both the descriptive attributes (i.e., independent variables or decision variables) and the class attribute (i.e., output variable or result variable). In contrast, with unsupervised learning, the training data includes only the descriptive attributes. Below picture shows a simple taxonomy for data mining tasks, along with the learning methods and popular algorithms for each of the data mining tasks. Out of the three main categories of tasks, prediction patterns/models can be classified as the outcome of a supervised learning procedure, while association and clustering patterns/models can be classified as the outcome of unsupervised learning procedures.



Prediction is commonly used to indicate telling about the future. It differs from simple guessing by taking into account the experiences, opinions, and other relevant information in conducting the task of foretelling. A term that is commonly associated with prediction is forecasting. Even though many people use these two terms synonymously, there is a subtle difference between them. Whereas prediction is largely experience and opinion based, forecasting is data and model based. That is, in the order of increasing reliability, one might list the relevant terms as guessing, predicting, and forecasting. In data mining terminology, prediction and forecasting are used synonymously, and the term prediction is used as the common representation of the act. Depending on the nature of what is being predicted, prediction can be named more specifically as classification (where the predicted thing, such as tomorrow's forecast, is a class label such as "rainy" or "sunny") or regression (where the predicted thing, such as tomorrow's temperature, is a real number, such as "65 degrees"). Classification, or supervised induction, is perhaps the most common of all data mining tasks. The objective of classification is to analyze the historical data stored in a database and automatically generate a model that can predict future behavior. This induced model consists of generalizations over the records of a training data set, which help distinguish predefined classes. The hope is that the model can then be used to predict the classes of other unclassified records and, more importantly, to accurately predict actual future events.

Common classification tools include neural networks and decision trees (from machine learning), logistic regression and discriminant analysis (from traditional statistics), and emerging tools such as rough sets, support vector machines, and genetic algorithms. Statistics-based classification techniques (e.g., logistic regression, discriminant analysis) have been criticized as making unrealistic assumptions about the data, such as independence and normality, which limit their use in classification-type data mining projects.

Neural networks involve the development of mathematical structures (somewhat resembling the biological neural networks in the human brain) that have the capability to learn from past experiences, presented in the form of well-structured data sets. They tend to be more effective when the number of variables involved is rather large and the relationships among them are complex and imprecise. Neural networks have disadvantages as well as advantages. For example, it is usually very difficult to provide a good rationale for the predictions made by a neural network. Also, neural networks tend to need considerable training. Unfortunately, the time needed for training tends to increase exponentially as the volume of data increases, and, in general, neural networks cannot be trained on very large databases. These and other factors have limited the applicability of neural networks in data-rich domains.

Decision trees classify data into a finite number of classes, based on the values of the input variables. Decision trees are essentially a hierarchy of if-then statements and are thus significantly faster than neural networks. They are most appropriate for categorical and interval data. Therefore, incorporating continuous variables into a decision tree framework requires discretization—that is, the conversion of continuous valued numeric variables to ranges and categories.

Using associations—which are commonly called association rules in data mining—is a popular and well researched technique for discovering interesting relationships among variables in large databases. Thanks to automated data-gathering technologies such as use of bar code scanners, the use of association rules for discovering regularities among products in large-scale transactions recorded by point-of-sale systems in supermarkets has become a common knowledge-discovery task in the retail industry. In the context of the retail industry, association rule mining is often called market-basket analysis.

Two commonly used derivatives of association rule mining are link analysis and sequence mining. With link analysis, the links among many objects of interest are discovered automatically, such as the link between web pages and referential relationships among groups of academic publication authors. With sequence mining, relationships are examined in terms of their order of occurrence to identify associations over time. Algorithms used in association rule mining include the popular Apriori (where frequent item sets are identified), FP-Growth, OneR, ZeroR, and Eclat algorithms.

Clustering involves partitioning a collection of things (e.g., objects, events, etc., presented in a structured data set) into segments (or natural groupings) whose members share similar characteristics. Unlike in classification, in clustering, the class labels are unknown. As the selected algorithm goes through the data set, identifying the commonalities of things based on their characteristics, the clusters are established. Because the clusters are determined using a heuristic-type algorithm, and because different algorithms may end up with different sets of clusters for the same data set, before the results of clustering techniques are put into use, it may be necessary for an expert to interpret and potentially modify the suggested clusters. After reasonable clusters have been identified, they can be used to classify and interpret new data.

Not surprisingly, clustering techniques include optimization. The goal of clustering is to create groups so that the members within each group have maximum similarity and the members across groups have minimum similarity. The most commonly used clustering techniques include k-means (from statistics) and self-organizing maps (from machine learning), which is a unique neural network architecture developed by Kohonen (1982).

Firms often effectively use their data mining systems to perform market segmentation with cluster analysis. Cluster analysis is a means of identifying classes of items so that items in a cluster have more in common with each other than with items in other clusters. This type of analysis can be used in segmenting customers and directing appropriate marketing products to the segments at the right time in

the right format at the right price. Cluster analysis is also used to identify natural groupings of events or objects so that a common set of characteristics of these groups can be identified to describe them.

Two techniques often associated with data mining are visualization and time-series forecasting. Visualization can be used in conjunction with other data mining techniques to gain a clearer understanding of underlying relationships. As the importance of visualization has increased in recent years, the term visual analytics has emerged. The idea is to combine analytics and visualization in a single environment for easier and faster knowledge creation.

In time-series forecasting, the data consists of values of the same variable that is captured and stored over time, at regular intervals. This data is then used to develop forecasting models to extrapolate the future values of the same variable.

Data Mining Applications

Data mining is widely used in diverse areas. There are a number of commercial data mining systems available today.

- Financial Data Analysis
- Retail Industry
- Telecommunication Industry
- Biological Data Analysis
- Other Scientific Applications
- Intrusion Detection
- Financial Data Analysis

The financial data in banking and financial industry is generally reliable and of high quality which facilitates systematic data analysis and data mining.

Some of the typical cases are as follows –

- Design and construction of data warehouses for multidimensional data analysis and data mining.
- Loan payment prediction and customer credit policy analysis.
- Classification and clustering of customers for targeted marketing.
- Detection of money laundering and other financial crimes.
- Retail Industry

Data Mining has its great application in Retail Industry because it collects large amount of data from on sales, customer purchasing history, goods transportation, consumption and services. It is natural that the quantity of data collected will continue to expand rapidly because of the increasing ease, availability and popularity of the web.

Data mining in retail industry helps in identifying customer buying patterns and trends that lead to improved quality of customer service and good customer retention and satisfaction.

Here is the list of examples of data mining in the retail industry –

Design and Construction of data warehouses based on the benefits of data mining.

Multidimensional analysis of sales, customers, products, time and region.

Analysis of effectiveness of sales campaigns.

- Customer Retention.
- Product recommendation and cross-referencing of items.
- Telecommunication Industry
- Today the telecommunication industry is one of the most emerging industries providing various services such as fax, pager, cellular phone, internet messenger, images, e-mail, web data transmission, etc. Due to the development of new computer and communication technologies, the telecommunication industry is rapidly expanding. This is the reason why data mining is become

very important to help and understand the business.

· Data mining in telecommunication industry helps in identifying the telecommunication patterns, catch fraudulent activities, make better use of resource, and improve quality of service. Here is the list of examples for which data mining improves telecommunication services –

- Multidimensional Analysis of Telecommunication data.

- Fraudulent pattern analysis.
- Identification of unusual patterns.
- Multidimensional association and sequential patterns analysis.
- Mobile Telecommunication services.
- Use of visualization tools in telecommunication data analysis.
- Biological Data Analysis

In recent times, we have seen a tremendous growth in the field of biology such as genomics, proteomics, functional Genomics and biomedical research. Biological data mining is a very important part of Bioinformatics. Following are the aspects in which data mining contributes for biological data analysis –

A Data Object

	Attributes				
	Tip	Refund	Marital Status	Taxable Income	Cheat
Objects	1	Yes	Single	125K	No
	2	No	Married	100K	No
	3	No	Single	70K	No
	4	Yes	Married	120K	No
	5	No	Divorced	95K	Yes
	6	No	Married	60K	No
	7	Yes	Divorced	220K	No
	8	No	Single	85K	Yes
	9	No	Married	75K	No
	10	No	Single	90K	Yes

- database rows → data objects
- database columns → attributes

Similarity and Dissimilarity

- Similarity
 - Numerical measure of how alike two data objects are.
 - Is higher when objects are more alike.
 - Often falls in the range [0,1]
- Dissimilarity
 - Numerical measure of how different are two data objects
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies
- Proximity refers to a similarity or dissimilarity

Similarity/Dissimilarity for Simple Attributes

p and q are the attribute values for two data objects.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal	$d = \frac{ p-q }{n-1}$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - \frac{ p-q }{n-1}$
Interval or Ratio	$d = p - q $	$s = \frac{1}{1+d}$ or $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$